

CO2 - Assessment Tool - 2

CLASS TEST

M. Kavya

192472370

CSA1610

DWDM

Q1(a) Define Data mining:

Data mining is the process of extracting useful patterns, relationships and Knowledge from large amounts of data using statistical & ML.

Any two Data mining Tasks:

- 1) classification
- 2) clustering

Steps in KDD (Knowledge Discovery in Databases)

- 1) Data selection
- 2) Data cleaning
- 3) Data Integration
- 4) Data transformation
- 5) Data mining
- 6) Pattern Evaluation
- 7) Knowledge presentation

Roles of Data processing:

- Removes missing & noisy data
- Improves data quality
- Increasing mining accuracy
- Reduces processing time.

Q1(b) Data cleaning:

Data cleaning is the process of detecting & correcting inaccurate, incomplete or inconsistent data.

95 / 100

JK

Two Data Quality Issues:

- Missing values & Noisy data

Handling missing values:

- Replace with mean/median.
- Replace using previous value or prediction.

Q2(a) Data Integration:

Data Integration is the process of combining data from diff sources into a single, consistent dataset.

challenges:

- Data redundancy (duplicate data)
- Data inconsistency

Handling:

- Redundancy: Remove duplicate records using Keys & normalization.
- Inconsistency: Standardize data formats to resolve conflicting values.

Q2(b) Covariance Analysis:

Covariance analysis measures how 2 variables change together

Two covariance formulas:

1) Population covariance:

$$\text{cov}(X, Y) = \frac{\sum (x - \mu_x)(y - \mu_y)}{N}$$

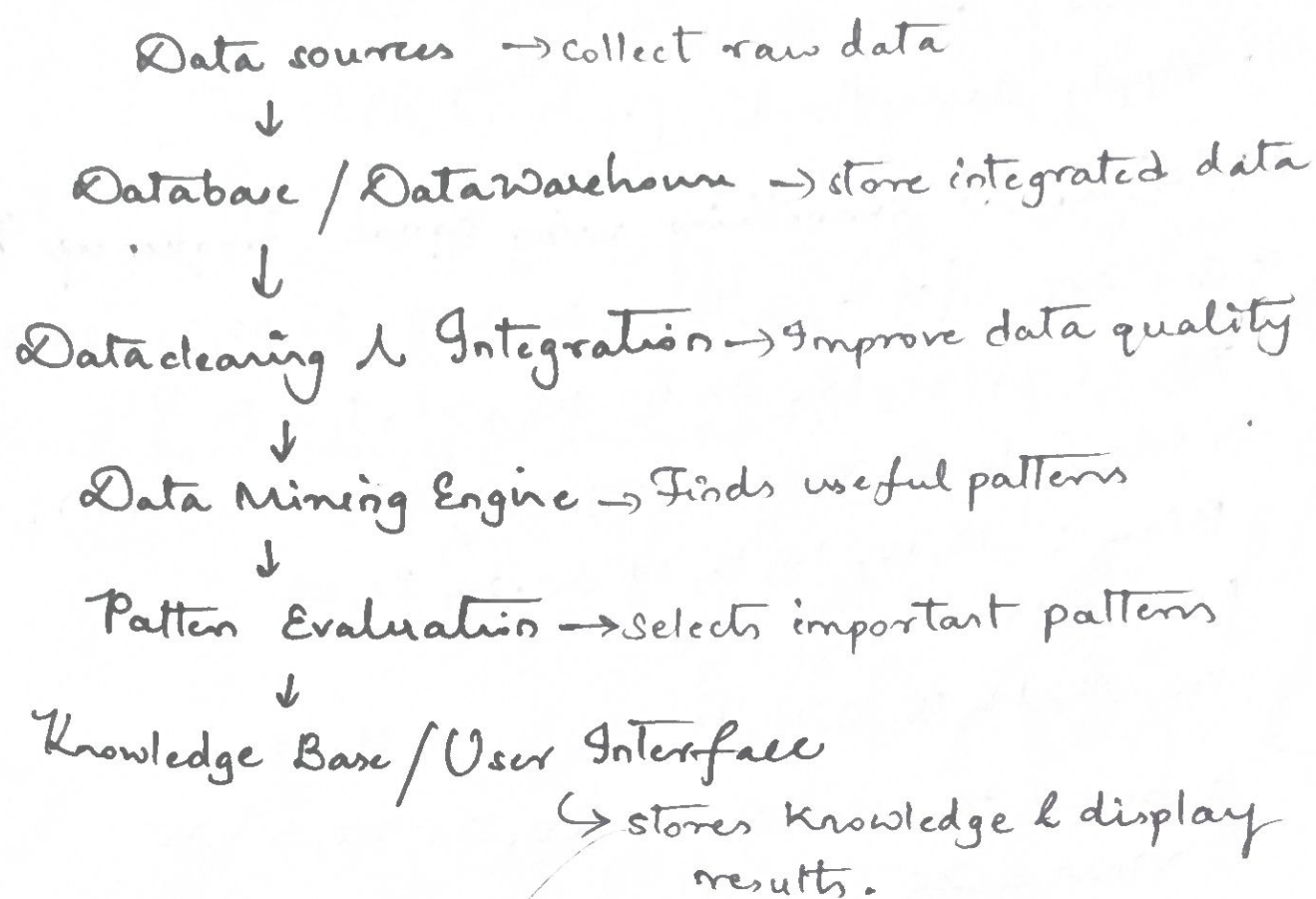
2) Sample covariance:

$$\text{cov}(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1}$$

Q3(a)

The values of stocks A & B increase together, indicating a positive covariance. Therefore, their prices rise together & they are not independent, as both are influenced by the same Industry Trends.

Q3(b) Architecture of a Data Mining system



Q4(a) Finding Covariance.

$$X = \{4, 5, 6, 7, 9\}; Y = \{8, 3, 7, 5, 6\}$$

$$\text{Mean}(X) = 6.2 \quad \text{Mean}(Y) = 5.8$$

$$\text{cov}(X, Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n-1} = \frac{-1.8}{4} = -0.45$$

$$\therefore \text{covariance} = -0.45$$

Q4(b) Data Pre-processing:

Major steps:

- Data cleaning
- Data Integration
- Data transformation
- Data Reduction
- Data Discretization

pre-processing pipeline:

- Handle missing values
- Remove outliers
- Standardize data format
- Transform/normalize data
- Reduce unnecessary data
- Apply discretization (if needed)

Q5(a) Local smoothing using Equal-Frequency Binning)

Given data: {13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70}

Divide into 3 equal-frequency (9 values each):

Bin-1: 13, 15, 16, 16, 19, 20, 20, 21, 22

Mean = 18

Smoothed: 18, 18, 18, 18, 18, 18, 18, 18, 18

Bin-2: 22, 25, 25, 25, 25, 30, 33, 33, 35

Mean = 26

Bin-3: 35, 35, 35, 36, 40, 45, 46, 52, 70

Mean = 44

Q5(b) : Data Discretization :

Data discretization is the process of converting continuous data into intervals.

Importance:

- Reduces data size
- Removes noise
- Improves Analysis & mining efficiency

Ex: 0-20, 21-40, 41-60

→ Each bin contains the same number of data values